Type MS · Protecting software by law: copyright, patents, trademarks and licences

Answer 1 of 2 · Question 6 part d · 6 marks

2021 NOV · Paper 01 · extract 1 of 6 · find part d in this mark scheme extract

6(a)(i)	Award **one** mark from: • faster translation/execution (1) • allows her to directly address components / make efficient use of available memory space (1) • can dispense with the need for an OS to make the code run faster/free up memory space (1)		1
6(a)(ii)	Award **one** mark from: • difficult to read/understand (1) • easy to make mistakes (1) • can be hard to find errors in the code (1) • time consuming (to write) (1) • not very portable / processor-specific (1) • lack of built-in functions/procedures (1) • few development/editorial tools available (1)		1

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Answer 1 of 2 · Question 6 part d · 6 marks

2021 NOV · Paper 01 · extract 2 of 6 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark from: • virus detection/scanner (1) • threat identification (1) • real-time scanning (1) • scan scheduling (1) • quarantine files (1) • carry out automatic updates (1) • on-demand file scanning (1) • kill switch (1)		2
6(b)(ii)	Award **one** mark from: • anti-spyware (1) • anti-adware (1)		1
6(c)	A AND O AND (W OR D) Award **one** mark each to a maximum of **four** marks for: • A AND O (1) • W OR D (1) • Brackets around W OR D (1) • AND between A AND O (W OR D) (1)		4

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Answer 1 of 2 · Question 6 part d · 6 marks

2021 NOV · Paper 01 · extract 3 of 6 · find part d in this mark scheme extract

6(d)	**Intellectual property** • Intellectual property is any work that is distinct, owned, and protected by patent or copyright laws • Akiko's software programs fall into this category • She will be able to protect her intellectual property • Her software is protected by copyright without her having to apply for it **Licensing** • Purchasing software does not mean you own it • Could add a licence key to the software. • Could require compulsory registration using the internet • Could allow proprietary licences, which do not allow code modification or code reuse. • Could allow Free and open-source software (FOSS), which would allow the user to modify and reuse the code. • Creative Commons (CC) ○ Could use creative commons (CC) licences that would allow the free distribution of copyrighted work ○ Would use if Akiko wanted to allow people to share, use and build on the programs. ○ May mention some of the CC licences and conditions	6

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Answer 1 of 2 · Question 6 part d · 6 marks

2021 NOV · Paper 01 · extract 4 of 6 · find part d in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.
		Total **15**

Type MS · Protecting software by law: copyright, patents, trademarks and licences

Answer 1 of 2 · Question 6 part d · 6 marks

2021 NOV · Paper 01 · extract 5 of 6 · find part d in this mark scheme extract

Type MS · Protecting software by law: copyright, patents, trademarks and licences

Answer 1 of 2 · Question 6 part d · 6 marks

2021 NOV · Paper 01 · extract 6 of 6 · find part d in this mark scheme extract

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Answer 2 of 2 · Question 5 part d · 3 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

5(a)	<table><tr><th>P</th><th>Q</th><th>R</th><th>Q OR R</th><th>P AND (Q OR R)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> Award **one** mark for each of: • All possible combinations in columns P, Q and R (1) • Correct values in Q or R (1) • Correct values in final column (1)	Allow follow through for incorrect mark point 2.	3

Question Number	Answer	Additional Guidance	Mark
5(b)	B AND S AND NOT L Award **one** mark for each of: NOT L or B AND S (1) Fully correct (1)	• Any order and ignore brackets	2

Question Number	Answer	Additional Guidance	Mark
5(c)	Any **two** of: • Remove code vulnerabilities in programming languages (1). • Eliminate bad programming practices (1).		2

Question Number	Answer	Additional Guidance	Mark
5(d)	Award **one** mark each for any of: • Copyright (1) • Patent (1) • Licensing (1) • Trademark (1)		3

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

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Answer 2 of 2 · Question 5 part d · 3 marks

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Answer 2 of 2 · Question 5 part d · 3 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark